Airborne P- and L-Band SAR Tomography for Forest Vertical Structure Mapping Using Only Four Images

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***Abstract*—Synthetic aperture radar (SAR) tomography (TomoSAR) technology effectively provides precise 3-D forest vertical structure (FVS) information. However, conventional TomoSAR methods require abundant acquisitions for accurate 3-D reconstruction, which is time-consuming and low-efficiency for FVS mapping. To address these limitations, this article proposes a novel microstack TomoSAR imaging approach utilizing four images, referred to as the double iterative adaptive residual approach (DIARA). The DIARA innovatively combines iterative inner–outer adaptive spectral estimation and residual optimization to enhance both processing efficiency and accuracy. For validation purposes, both simulated and airborne TomoSAR experiments are analyzed at P and L bands. The P-band results from the BorTomoSAR campaign indicate that the DIARA acquires higher accuracy for estimating forest height than the conventional methods, i.e., improving R^2 from 0.443 to 0.628, mean absolute error (MAE) from 0.779 to 0.625 m, mean absolute percentage error (MAPE) from 4.629% to 3.680%, and root mean square error (RMSE) from 0.941 to 0.771 m. In addition, the performance is further validated by the TropiSAR P-band campaign, which confirms the robustness of the proposed DIARA in high-canopy, complex forest environments. Furthermore, the L-band results from HaiTomoSAR campaign demonstrate that the DIARA method successfully detects the weak ground scatterers beneath dense forest canopies, which validates its super-resolution capability in vertical structure reconstruction.**

***Index Terms*—3-D forest mapping, forest vertical structure (FVS), L band, P band, synthetic aperture radar tomography (TomoSAR).**

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I. Introduction

A. Background

FORESTS, as the largest terrestrial ecosystems, play critical roles in regulating global carbon cycles, biodiversity, and climate stability [1], [2], [3]. The accurate characterization of forest vertical structure (FVS) is essential for reliably estimating biomass, ecosystem productivity, and carbon stocks, thereby directly supporting climate change research and informing carbon policy [4], [5]. Airborne synthetic aperture radar (SAR) tomography (TomoSAR) provides unique capabilities to reconstruct 3-D forest structures through multibaseline SAR acquisitions [6], [7]. The efficacy of TomoSAR for forest mapping has been extensively validated at P and L bands across diverse forest ecosystems [8], [9], including temperate [10], [11], boreal [12], [13], and tropical forests [14], [15], [16]. Given these proven capabilities, airborne TomoSAR has become an essential platform validating mission concepts and refining algorithms for upcoming spaceborne SAR missions, such as Chinese LuTan-1 [17], ESA's BIOMASS [18], and DLR's Tandem-L [19].

High-quality TomoSAR imaging requires multiple SAR acquisitions with uniformly distributed baselines along the elevation direction to ensure reliable vertical resolution [20], [21]. However, imaging performance is significantly degraded when the number of acquisitions is limited or the baseline geometry is irregular [6]. Moreover, the requirement for a large TomoSAR image stack (TIS) substantially increases mission costs, flight durations, and data volume, significantly limiting the operational feasibility and efficiency of the technology. For TomoSAR applications at longer wavelengths (e.g., P and L bands), long spatial baselines are typically employed. This, however, contributes significantly to spatial and temporal decorrelation, consequently diminishing interferometric coherence and degrading the quality of 3-D reconstruction [22], [23]. Therefore, improving the tomographic imaging performance under limited acquisition conditions is essential to enhance the practicality for forest monitoring applications.

B. State of the Art

In recent years, various spectral estimation techniques have been extensively investigated to improve tomographic focusing

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performance. They are generally divided into nonparametric, parametric, and sparse spectral estimation methods [7].

1) Nonparametric Spectral Estimation: The beamforming is the most classical and widely adopted technique due to its implementation simplicity and robustness, which has been successfully applied in multiple campaigns, including BioSAR [13], TempoSAR [24], TropiSAR [25], AfriSAR [26], and TomoSense [27]. However, the beamforming method is inherently limited by Rayleigh resolution and lacks super-resolution capability. To overcome the above limitation, adaptive spectral estimation methods have been introduced, which attenuate interfering scattering components based on the design of spatial filters to suppress interference from undesired elevation layers. Therein, the well-known Capon spectral estimator proposed by Lombardini and Reigber [28] is the most representative method, which improves resolution beyond the Rayleigh limit. Despite its improved performance, Capon remains sensitive to noise and prone to sidelobe artifacts. To further enhance resolution and robustness, the iterative adaptive approach (IAA), based on weighted least squares, has been successfully developed [29]. However, particularly in airborne TomoSAR systems with limited tomographic apertures, IAA is susceptible to performance degradation and unreliable estimates when its covariance matrix becomes ill-conditioned. Therefore, two improved versions of IAA, i.e., IAA-ML [30] and RIAA [31], are designed to enhance resolution and estimation robustness in forest scenes. The above methods maintain relatively high computational efficiency and are feasible for engineering applications. Nevertheless, uneven baseline distributions degrade resolution, introduce artifacts, and increase ambiguity.

Starting from the enhanced TomoSAR Imaging, the MARIA method proposed by Martín del Campo et al. [32] combines a maximum-likelihood framework with an iterative refinement strategy, which suppresses artifacts, and reduces ambiguity. In 2020, Martín del Campo et al. [33] further relaxed and developed the probability density function of the observed data. They integrated the weighted covariance fitting (WCF) criterion with a statistical regularization framework to develop the WCF-based iterative spectral estimator (WISE), which has demonstrated improved robustness in both forest and urban scenarios. However, the method remains sensitive to initial power estimates. Wang et al. [34] proposed the G-Pisarenko method, which applies logarithmic transformation and eigenvalue weighting to the covariance matrix. This method effectively enhances dominant scattering components and reduces sidelobe levels. However, it imposes a high computational burden due to repeated matrix operations and eigenvalue decomposition steps.

2) Parametric Spectral Estimation: The parametric estimation methods decompose the observed TIS signal into signal and noise subspaces based on the eigenvalue analysis, which have been widely explored in TomoSAR to enhance the vertical resolution and extract structural features from volumetric forest scenes. Nannini et al. [35] investigated the minimum number of acquisitions required for 3-D reconstruction using multiple signal classification (MUSIC) method. The MUSIC method effectively separates the signal and noise subspaces through eigenstructure decomposition, and achieves super-resolution by projecting steering vectors onto the noise subspace. Huang et al. [36] proposed signal subspace fitting (SSF) and noise subspace fitting (NSF) methods for polarimetric TomoSAR, which suppress speckle, enhance under-foliage target detection, and improve elevation estimation.

In recent years, regularization strategies have integrated regularization strategies into TomoSAR imaging, allowing adaptive control over sidelobe suppression, sparsity enforcement, and model complexity through tunable parameters. Lu et al. [37] proposed a modified Capon method (M-Capon), which enhances resolution by iteratively removing dominant scatterers and reduces interference between ground and canopy components. Haddad et al. [38] introduced a generalized parametric iterative approach (GPIA), which unifies several ML-based iterative method through a regularization parameter. Bou et al. [39] developed a low-dimensional parametric model using P-band single-polarization data for tropical forest characterization. However, the simplification may limit its performance in heterogeneous or multilayered canopies.

3) Sparse Spectral Estimation: The sparse spectrum estimation methods have emerged as a powerful approach in TomoSAR for reconstructing FVS. These methods reformulate the TomoSAR imaging problem as an optimization task to achieve super-resolution performance. Since 2013, based on the sparsity of forest reflectivity in the wavelet domain, Aguilera et al. [40] first introduced the concept of compressed sensing (CS) to achieve high resolution and attain low ambiguity levels. Li et al. [41] combined a sum of Kronecker product (SKP) decomposition with CS to reconstruct the reflectivity signals of ground and canopy scattering. Huang et al. [42] proposed a wavelet-based hybrid estimator that simultaneously captured sparse object responses and distributed volume scattering. Martín del Campo et al. [43] introduced a multistage strategy combining virtual adaptive beamforming and wavelet-domain sparsity refinement to enhance vertical feature resolutions. However, the effectiveness of these methods are sensitive to both the chosen optimization framework and the configuration of model parameters. Moreover, the reliance on iterative procedures and matrix computations imposes substantial computational demands. Peng et al. [44] proposed a SPICE-based sparse spectral estimator that leverages wavelet and orthogonal basis to adaptively recover forest structure with minimal acquisitions and without the need for hyperparameter tuning. Wan et al. [45] proposed a covariance vector sparse bayesian learning method, which formulates the tomographic inversion as a bayesian estimation problem in the wavelet domain.

Multipass TomoSAR systems, especially at P and L bands, often rely on long baselines to achieve sufficient spatial baseline span. However, such configurations are highly susceptible to spatial and temporal decorrelation over forested areas, resulting in diminished interferometric coherence and compromised 3-D reconstruction [46]. To mitigate these effects, microstack configurations with only four acquisitions have gained attention due to their reduced data requirements and increased resilience to temporal variability. Nevertheless, conventional spectral estimation methods are primarily designed

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for large tomographic image stacks and often fail to preserve resolution and suppress sidelobes under limited acquisition conditions. There is a growing need for spectral estimation method that can operate reliably under limited acquisition conditions and provide super-resolution tomographic results.

C. Solution and Contributions

This article proposes a novel tomographic SAR imaging approach termed the double iterative adaptive residual approach (DIARA), which is specifically designed to overcome current limitations in microstack TomoSAR. Within the first step, an inner iterative is established to resolve underdetermined linear systems and realize tomographic focusing. Within the second step, a residual optimization model is designed to improve the estimation accuracy and mitigate sidelobe. Within the third step, an outer iterative thresholding strategy is employed to refine tomographic focusing. Compared with existing approaches, the main contributions of this article are summarized as follows.

1) The fundamental challenges of microstack TomoSAR are systematically analyzed, including the underdetermined nature of the TomoSAR inversion problem, the degradation of elevation resolution caused by small tomographic aperture, and the difficulty of accurate 3-D reconstruction under sparse baseline sampling. In forested environments, characterized by the presence of both ground and canopy scatterers, resolving the two dominant scattering phase centers (SPCs) typically requires a minimum of four SAR acquisitions.
2) A comprehensive reconstruction method named DIARA is proposed to overcome the aforementioned limitations. First, an inner iterative procedure is designed to enhance the tomographic focusing by refining the residual spectrum adaptively, achieving accurate localization of dominant scatterers. Second, a residual-based optimization model is introduced to attenuate sidelobe energy and improve image clarity. Third, an outer iterative refinement further sharpens the tomographic profile. By integrating tomographic profile refinement and residual-driven enhancement, DIARA enables robust and high-resolution 3-D imaging under limited TIS.
3) Extensive experiments are conducted using both simulated and airborne TomoSAR experiments assess the effectiveness of DIARA. The simulated P-band SAR data are designed to evaluate the accuracy, robustness, and super-resolution capability. In addition, three airborne TomoSAR stacks, including P-band BorTomoSAR campaign, a challenging P-band TropiSAR campaign, and L-band HaiTomoSAR campaign, are employed to test the super-resolution capability and generalizability of the proposed DIARA. The acquired TomoSAR data are employed to test super-resolution capability of the proposed DIARA. Qualitative tomographic profiles and quantitative metrics with respect to LiDAR data are conducted and analyzed. The results demonstrate significant improvements in 3-D reconstruction quality and applicability under microstack constraints.

D. Organization

The remainder of this article is organized as follows. Section II provides a background on TomoSAR principles and formulates the reconstruction problem. The proposed methodology is detailed in Section III, followed by numerical evaluations using both simulated and airborne P- and L-band TIS in Section IV. Finally, Section V concludes this article with a summary of key findings.

II. Models and Problem Statement

A. Principles of TomoSAR Imaging

The fundamental principle of tomographic SAR involves acquiring data from multiple tracks over the area of interest, which are typically parallel to each other. The acquisition creates a synthetic aperture in the elevation direction, enabling the reconstruction of the signal through a 2-D synthetic aperture array and achieving 3-D resolution [47]. Considering the acquisition geometry of TomoSAR, N single-look complex (SLC) SAR images are obtained along N multiple parallel tracks, as shown in Fig. 1. The TIS is acquired after preprocessing steps, including co-registration, topographic phase removal, and phase calibration concerning a common primary track. Consequently, for a specific azimuth–range pixel (with a- and r-coordinates), and its corresponding multibaseline complex observations, the standard model of the nth element in the TIS vector, $\mathbf{g} \in \mathbb{C}^{N\times 1}$, is presented as follows [48]:

$$g_n(a,r) = \int_{\Delta z} \gamma(a,r,z) \exp(jK_n z)\, \mathrm{d}z \tag{1}$$

where $\gamma(a,r,z)$ denotes the reflectivity function at position (a,r,z) and Δz denotes the vertical extent in the elevation dimension. The wavenumber in the elevation direction for the nth image is defined as $K_n = (4\pi b_n/\lambda R \sin(\theta))$, where b_n denotes the component of the spatial perpendicular baseline relative to the primary track, θ is the local incidence angle, and the λ represents the operating wavelength of the radar sensor. The reflectivity distribution along elevation is approximated by a discrete representation, and random measurement perturbations $\boldsymbol{\varepsilon}$ are included in the model. Then, the corresponding discrete formulation is expressed as follows:

$$\mathbf{g} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} \tag{2}$$

where $\mathbf{g} = [g_1, g_2, \ldots, g_N]^{\mathrm{T}}$ is the complex measurement vector with N elements. The matrix $\mathbf{A} \in \mathbb{C}^{N\times M}$ is the mapping matrix, whose elements are defined as $A_{n,m} = \exp(jK_n z_m)$, where z_m represents elevation sampling values. $\boldsymbol{\gamma} = \left[\gamma(z_1), \gamma(z_2), \ldots, \gamma(z_M)\right]^{\mathrm{T}}$ is an $M \times 1$ discrete reflectivity vector, where each entry $\gamma_m = \gamma(z_m)$ corresponds to the reflectivity at the mth elevation sampling point z_m.

B. Parametric Estimation Problem

The reconstruction of 3-D FVS relies on a statistical scattering model. Specifically, TomoSAR imaging is employed to estimate the vertical profile of backscattered power. In forested environments, the received radar signal typically arises from four dominant mechanisms: direct ground scattering, trunk-ground double-bounce scattering, canopy–ground

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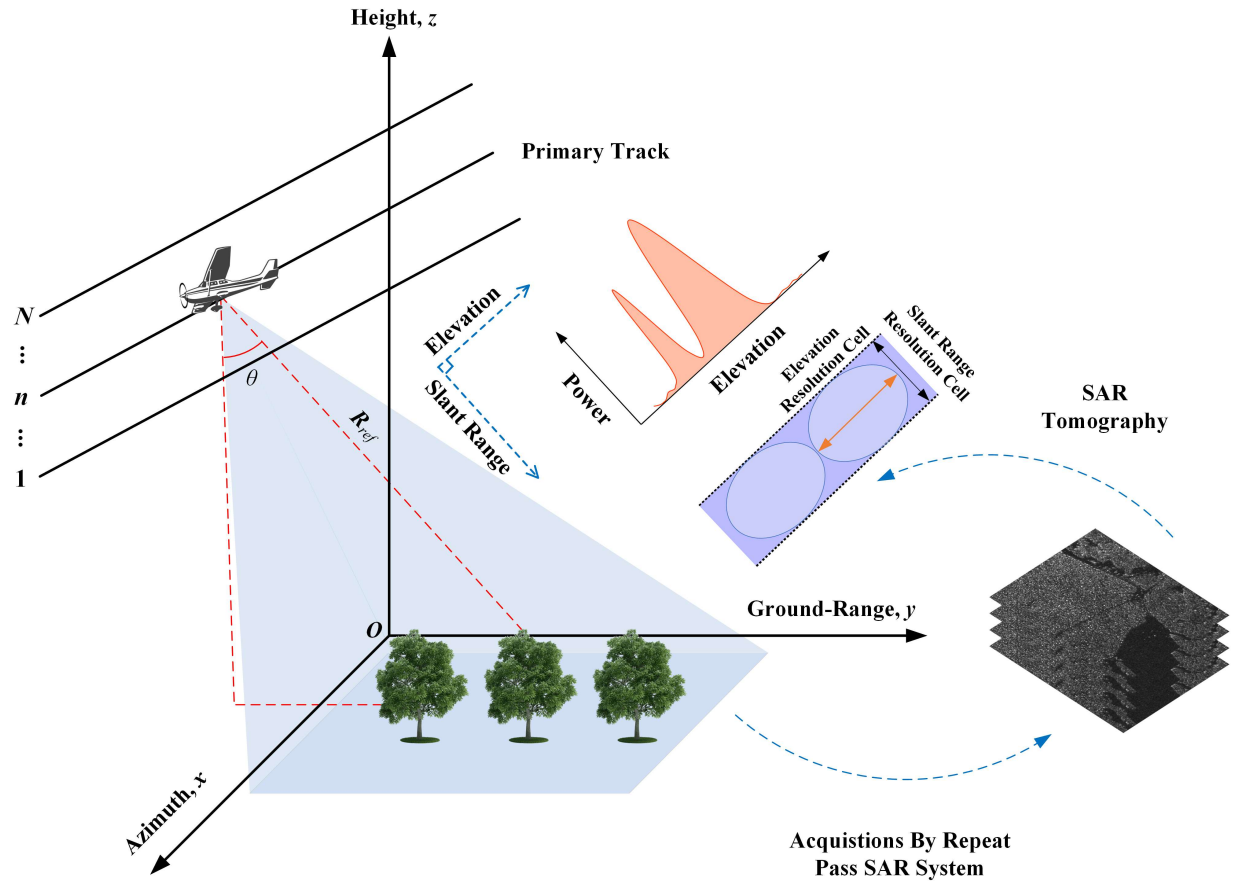


Fig. 1. Flowchart of 3-D reconstruction in TomoSAR using a repeat-pass SAR system.

double-bounce scattering, and volume scattering within the canopy layer [12], [48], as shown in Fig. 2. Overall, the forest scene is characterized by the superposition of effective SPCs linked to returns from both the ground and the canopy along the elevation direction. Suppose that $\gamma(s)$ follows a Gaussian distribution and is a white random signal. The expected value of interferometric signal facilitates a cohesive mathematical approach to handle various scattering mechanisms previously outlined, which is formulated as follows:

$$E\left\{g_m g_n^*\right\} = \int_{\Delta z} p\left(z\right) \exp\left[j\left(K_m - K_n\right) z\right] \mathrm{d}z \tag{3}$$

where $p(z)$ represents the spatial power spectrum. K_m and K_n are the elevation wavenumbers associated with acquisitions m and n, respectively. Assuming that there are K contributing effective SPCs, to facilitate a practical representation and incorporate noise power p_0 contributions, the model in (3) is approximated in discrete form as follows:

$$E\left\{g_m g_n^*\right\} = \sum_{k=1}^{K} p_k \gamma_{mn}^{(k)} + p_0 \delta_{mn} \tag{4}$$

where p_k denotes the power spectrum contribution from the kth SPC, γ_{mn}^k represents the complex interferometric coherence associated with the spatial structure of the kth SPC [12], and δ_{mn} is the Kronecker delta which accounts for the additive noise power on the interferometric signal. The complex coherence term γ_{mn}^k is described by a parametric model that incorporates both decorrelation effects and phase delays arising from elevation differences. The amplitudes of the coherence is modeled as follows:

$$|\gamma_{mn}^k| = \rho_k^{-|b_m - b_n|/\Delta b} \tag{5}$$

and the corresponding phase is given by

$$\arg\left\{\gamma_{mn}^k\right\} = \left(K_m - K_n\right) z_k \tag{6}$$

where $\rho_k \in [0, 1]$ is the decorrelation factor that characterizes the angular spreading of the signal associated with the kth SPC, Δb is the average sampling interval along the normal baseline direction, and z_k represents the elevation of the kth SPCs. A decorrelation factor close to one indicates high coherence, lower values reflect volumetric scattering from the canopy.

In TomoSAR imaging for forest monitoring, retrieving vertical structural parameters such as forest height typically involves estimating the elevation positions of the dominant SPCs, which are often associated with the ground and the canopy layers within the resolution cell. This estimation process is based on the statistical characterization of the TIS, which is represented through its covariance matrix. Under the K SPCs model, the covariance matrix takes the form

$$C = E\left\{\mathbf{g}\mathbf{g}^{\mathrm{H}}\right\} = \sum_{k=1}^{K} p_k \Gamma_k + p_0 \mathbf{I} \tag{7}$$

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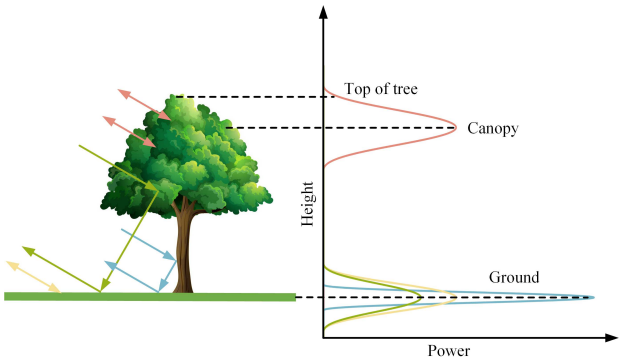


Fig. 2. Schematic of forest vertical reflectivity distribution. The vertical distribution of the forest backscattering mechanism power spectrum. Blue: trunk-ground scattering; yellow: ground backscattering; pink: canopy backscattering; and green: canopy–ground scattering.

where $(\cdot)^{\mathrm{H}}$ is the Hermitian transpose operation. $\mathbf{g} \in \mathbb{C}^N$ denotes the vector of SLC observations, Γ_k is an $N \times N$ the structure matrix associated with the kth SPCs, and I is the identity matrix.

In model (7), the total set of unknown parameters includes the power coefficients p_k, the elevation positions z_k, the spreading contants ρ_k, and the noise power p_0, resulting in a total of $3K + 1$ real-valued parameters. The number of effective scatterers K is generally unknown and is typically estimated from the data using model order selection strategies such as minimum description length (MDL) [49]. In airborne TomoSAR systems, the use of uniformly spaced baselines typically results in a covariance matrix that approximates a Toeplitz structure. Under this assumption, only the upper triangular part of the Hermitian matrix is used in order to avoid redundancy. Each complex-valued measurement provides two real-valued equations corresponding to its real and imaginary components. Consequently, the overall count of independent measurements is $N_p = 2(N-1)+1$. A necessary condition for estimating a model with K SPCs is that the number of independent real-valued observations exceeds the total number of unknown parameters. The minimum requirement is given by

$$N_{\min} = \frac{3}{2}K + 1. \tag{8}$$

In forest environments, the vertical scattering profile is often dominated by two effective phase centers corresponding to the ground and the canopy. Substituting $K = 2$ into the theoretical requirement yields $N_{\min} = 4$, which establishes the minimum number of acquisitions needed for reliable tomographic reconstruction under this model assumption.

III. Methodology

To achieve the 3-D reconstruction of the forest structure, it is essential to resolve the underdetermined linear system formulated in (2). As described in Section II, the available four SAR acquisitions constitute the theoretical minimum necessary for separating two dominant SPCs in forested areas. Fig. 3 presents the complete TomoSAR reconstruction chain, comprising co-registration, topographic phase removal, NC-PGA phase calibration of the microstack [9], and DIARA 3-D

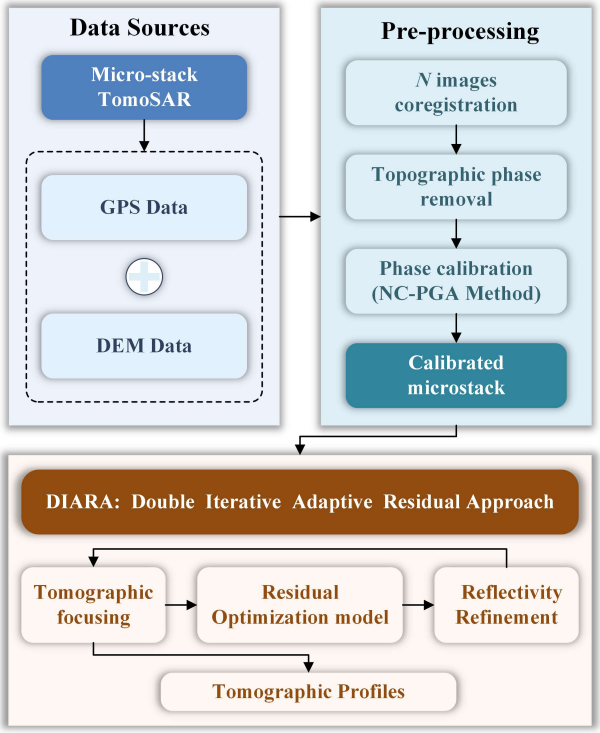


Fig. 3. Flowchart of the microstack TomoSAR reconstruction.

imaging. Importantly, the identical four-image microstack is employed throughout every stage, obviating the need for any additional data acquisition. Therefore, the 3-D reconstruction of forest structures is confronted with three problems, i.e., the underdetermined linear system of the tomographic focusing, the limited number of SAR acquisitions, and the degraded elevation resolution caused by a narrow elevation aperture. To address the above issues, a novel reconstruction framework, termed the DIARA, is proposed. The DIARA framework comprises three main steps: 1) tomographic focusing via an inner iterative approach; 2) sidelobe suppression via a residual-based model; and 3) tomographic profile refinement via a double iterative strategy.

A. Tomographic Focusing

Tomographic focusing aims to recover the vertical distribution of ground and canopy scatterers. This process involves solving an underdetermined linear system, which can be viewed as a 1-D spectral estimation problem [50]. An inner iterative approach is employed to enhance signal focusing along the elevation direction. To formulate the tomographic focusing problem, a linear measurement model is constructed to describe the mapping between elevation positions and observed SAR signals. The formulation begins with the linear measurement model described in (2), where the mapping matrix $\mathbf{A}$ can be rewritten as follows:

$$\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_M). \tag{9}$$

Each column $\mathbf{a}_m \in \mathbb{C}^N$ is defined as follows:

$$\mathbf{a}_m = \left[\exp\left(\mathrm{j}\mathrm{K}_1 z_m\right), \exp\left(\mathrm{j}\mathrm{K}_2 z_m\right), \ldots, \exp\left(\mathrm{j}\mathrm{K}_N z_m\right)\right]^{\mathrm{T}} \tag{10}$$

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where K_n denotes the spatial wavenumber associated with the nth acquisition and z_m is the mth discrete elevation sampling position.

The diagonal matrix $\mathbf{P} \in \mathbb{R}^{M\times M}$ represents the spatial power distribution along elevation and is denoted by

$$\hat{\mathbf{P}} = \left[|\gamma(z_1)|^2, |\gamma(z_2)|^2, \ldots, |\gamma(z_M)|^2\right]^{\mathrm{T}} \tag{11}$$

which approximates the backscattered power spectrum at each elevation position. To reduce spectral estimation variance and improve robustness, the spatial smoothing is applied over a local window of L samples. This technique reduces variance by averaging local realizations, which is particularly beneficial in the presence of limited observations and noise. It is widely used in high-resolution spectral estimation. The resulting power spectrum estimator is given as follows:

$$\hat{\mathbf{P}} = \frac{1}{L}\sum_{l=1}^{L}\left|\frac{\mathrm{diag}\left(\mathbf{A}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{g}(l)\right)}{\mathbf{A}^{\mathrm{H}}\mathbf{R}^{-1}\mathbf{A}}\right|^2 \tag{12}$$

where $\mathbf{g}(l)$ denotes the lth sample within the smoothing window and $\mathbf{R}$ is the data covariance matrix, which is initialized as the identity matrix $\mathbf{I}$ in the first iteration. This initialization is equivalent to conventional Fourier beamforming [6]. While this initialization provides a coarse estimate of the elevation power distribution, it lacks the resolution and sidelobe suppression required for accurate tomographic reconstruction under limited-baseline conditions. Therefore, an iterative refinement strategy is introduced to progressively enhance the focusing performance. This strategy progressively improves the tomographic reconstruction by incorporating updated information from the data covariance matrix at each iteration.

Convergence is monitored using the relative change in the ℓ_1 norm of the estimated power spectrum between successive iterations. The use of the ℓ_1-norm reflects the sparsity characteristic of the elevation scattering profile, where only ground and canopy scatterers contribute significantly to the total backscattered power. As a result, stabilization of the ℓ_1-norm serves as an effective indicator for convergence. Compared with energy-based norms such as the ℓ_2-norm, the ℓ_1-norm is more responsive to variations in the distribution of scattering power. It is therefore more suitable for monitoring convergence when the energy of tomographic profile is primarily concentrated in distinct regions, such as the ground and canopy layers in forested areas. The iteration proceeds until the relative difference between two consecutive ℓ_1-norm values falls below a predefined threshold T_f. Once this condition is satisfied, the estimated power spectrum is considered stable, and the iterative process is terminated. Therefore, the convergence condition is formally defined as follows:

$$\frac{\left\|\hat{\mathbf{P}}^{(i)} - \hat{\mathbf{P}}^{(i-1)}\right\|_{l_1}}{\left\|\hat{\mathbf{P}}^{(i-1)}\right\|_{l_1}} < T_f \tag{13}$$

where i is the iteration index and T_f is an empirically chosen threshold (set to 0.0001 in this study). The l_1 norm is computed as follows:

$$\left\|\hat{\mathbf{P}}\right\|_{l_1} = \sum_{m=1}^{M}\sqrt{\hat{P}_m} \tag{14}$$

where $\hat{P}_m = |\gamma(z_m)|^2$. During each iteration, the data covariance matrix $\mathbf{R}$ is updated based on the current estimate of $\hat{\mathbf{P}}$, according to the following expression:

$$\mathbf{R} = \mathbf{A}\mathbf{P}\mathbf{A}^H \tag{15}$$

where $\mathbf{P} = \mathrm{diag}(\hat{\mathbf{P}})$. This adaptive update ensures that the covariance structure reflects the evolving spectral distribution, allowing the focusing process to better emphasize dominant scatterers while suppressing sidelobes and noise.

B. Residual Optimization Model

To improve the accuracy of scatterer separation in tomographic reconstruction, a residual signal optimization strategy is proposed as the second step of the DIARA framework. The nonlinear, subtractive process aims to isolate individual scattering contributions by iteratively removing the influence of other phase centers. The core principle of this process is to address the multiple scattering contribution by sequentially isolating and characterizing each scattering contribution. First, the strongest scattering component within the resolution cell is identified and its parameters are estimated. Based on these estimates, its contribution is analytically modeled and then subtracted from the original measured signal, yielding a residual signal. This residual signal, now free from the influence of the dominant scatterer, is then used to locate and characterize the next strongest component.

The data covariance matrix is estimated through spatial averaging of complex-valued measurement vectors, following the principles of signal subspace decomposition. Assuming that an azimuth–range resolution cell contains K_{spc} effective SPCs, the observed signal $\mathbf{g}(l)$ for the lth spatial sample can be modeled as follows:

$$\mathbf{g}(l)_n = \sum_{k=1}^{K_{\mathrm{spc}}}\gamma(l)_k \exp(jK_n z_k) \tag{16}$$

where s_k is the elevation of the kth scatterer and $\gamma(l)_k$ is its corresponding complex-valued reflectivity. This expression models the observed signal as a coherent superposition of multiple SPCs, each with distinct elevation and reflectivity.

The number of effective scatterers K_{spc} is generally unknown and is typically estimated from the data using model order selection strategies [36]. Given initial estimates of the scatterer elevations and reflectivities $\{z_k, \gamma_k\}_{k=1}^{K_{\mathrm{spc}}}$, the contribution of the kth scatterer can be extracted by subtracting the estimated interference from all other components

$$\mathbf{g}(l)_k = \mathbf{g}(l) - \sum_{t=1,t\neq k}^{K_{\mathrm{spc}}}\gamma(l)_t\,\mathbf{a}(z_k). \tag{17}$$

This subtraction yields a residual signal that is primarily associated with the kth scatterer. Based on this isolated signal, the corresponding power spectrum is independently estimated using a spectral estimator as follows:

$$\hat{\mathbf{P}}_k = \frac{1}{L}\sum_{l=1}^{L}\left|\frac{\mathrm{diag}\left(\mathbf{A}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{g}(l)_k\right)}{\mathbf{A}^{\mathrm{H}}\mathbf{R}^{-1}\mathbf{A}}\right|^2. \tag{18}$$

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The elevation corresponding to the peak spectral energy is selected as the updated location of the kth phase center

$$\hat{z}_k = \arg\max \left\{ \hat{\mathbf{P}}_k \right\}. \tag{19}$$

Once the elevation is determined, the reflectivity of the kth component is updated via a least-squares solution

$$\hat{\gamma}(l)_k = \left[\mathbf{a}^{\mathrm{H}}(z_k)\, \mathbf{a}(z_k) \right]^{-1} \mathbf{a}^{\mathrm{H}}(z_k)\, \mathbf{g}(l)_k. \tag{20}$$

This iterative subtraction and estimation process is repeated for each scatterer, ensuring that each component's contribution is precisely isolated from the others. In each iteration, the signal model is refined using the most recent parameter estimates to ensure that the subtraction accurately removes the targeted scatterer's contribution. The processing procedure is crucial for achieving a precise separation of scatterers, effectively mitigating the interference caused by vertical layover between the ground and canopy layers in a forested environment.

C. Reflectivity Refinement via Iterative Thresholding

The third step of the DIARA framework employs an iterative thresholding strategy to jointly refine the elevation and reflectivity estimates of each SPC. This step complements the residual optimization process and is designed to enhance both the robustness and accuracy of scatterer separation, particularly in challenging forest environments with overlapping scattering responses.

The procedure begins by initializing the residual signal as $\mathbf{g}(l)_1 = \mathbf{g}(l)$, and assuming the first scatterer index $k = 1$. Based on this initialization, the elevation and reflectivity of the first scatterer are estimated using the previously described spectral estimation and least-squares procedures. The resulting parameters $\{z_1, \gamma(l)_1\}$ are then used to compute the updated residual signal $\mathbf{g}(l)_2$ by subtracting the estimated contribution of the first scatterer. The second component is then extracted from the updated residual using the same estimation process, yielding a new pair of parameters $\{z_2, \gamma(l)_2\}$. This alternating refinement between elevation and reflectivity estimates for each scatterer is repeated iteratively. At each iteration, the residual signal is updated and the parameter estimates are recalculated until convergence is achieved.

Convergence is determined by evaluating a cost function that measures the average residual energy across all samples in the smoothing window. Specifically, the cost is defined as the mean squared error between the observed signal and the reconstructed signal composed of all estimated components. This cost function reflects how well the current estimates explain the observed data, which is expressed as follows:

$$\mu_c = \frac{1}{L} \sum_{l=1}^{L} \left\| \mathbf{g}(l) - \sum_{k=1}^{K_{\mathrm{spc}}} \gamma(l)_k\, \mathbf{a}(z_k) \right\|_2^2 \tag{21}$$

where c denotes the current iteration index. The stopping condition is based on the relative change in the cost function between two successive iterations

$$|\mu_c - \mu_{c-1}| \le \zeta. \tag{22}$$

The iterative process terminates once the change rate in the optimization objective across iterations is lower than a designated threshold ζ. A smaller value of ζ typically leads to higher reconstruction accuracy, but it demands greater processing resources. In this work, ζ is empirically set to 0.0001 to achieve a balance between performance and efficiency. This iterative thresholding mechanism is particularly effective in stabilizing elevation and amplitude estimates when multiple scattering mechanisms occur within a single resolution cell. It also mitigates the influence of high sidelobes or closely spaced scatterers, which are common in forest scenes with vertical layover. To clarify the relationship among tomographic focusing, residual optimization, and thresholded refinement, the complete DIARA processing pipeline is summarized in Algorithm 1.

Algorithm 1 DIARA for Forest TomoSAR

Input: Tomographic stack $\{\mathbf{g}(l)\}_{l=1}^{L}$, mapping matrix $\mathbf{A}$, smoothing factor L, number of scatterers $K_{spc} = 2$, thresholds T_f, ζ

  Initialization: $\mathbf{g}(l)_1 = \mathbf{g}(l)$, $\mathbf{R} = \mathbf{I}_{N\times N}$, $k = 1$, $i = 0$, $c = 0$

  Repeat

    Estimate $\hat{\mathbf{P}}_k^i = \frac{1}{L}\sum_{l=1}^{L} |\mathrm{diag}(\mathbf{A}^{\mathrm{H}}\mathbf{R}^{-1}\mathbf{g}(l)_k / \mathbf{A}^{\mathrm{H}}\mathbf{R}^{-1}\mathbf{A})|^2$

    Update $\mathbf{P}_k = \mathrm{diag}(\hat{\mathbf{P}}_k^i)$, $\mathbf{R} = \mathbf{A}\mathbf{P}_k\mathbf{A}^{\mathrm{H}}$

    $i \leftarrow i + 1$

  until $\frac{\|\hat{\mathbf{P}}_k^i - \hat{\mathbf{P}}_k^{i-1}\|_1}{\|\hat{\mathbf{P}}_k^{i-1}\|_1} \le T_f$

  Repeat

    for $k = 1$ to K_{spc} **do**

      if $k > 1$ **then**

        $\mathbf{g}(l)_k = \mathbf{g}(l) - \sum_{t=1, t\neq k}^{K_{spc}} \gamma(l)_t\, \mathbf{a}(z_k)$

      end if

      Estimate $\hat{P}_k$, then $\hat{z}_k = \arg\max \hat{P}_k$

      $\hat{\gamma}(l)_k = [\mathbf{a}^{\mathrm{H}}(\hat{z}_k)\mathbf{a}(\hat{z}_k)]^{-1}\mathbf{a}^{\mathrm{H}}(\hat{z}_k)\mathbf{g}(l)_k$

    end for

    Compute $\mu_c = \frac{1}{L}\sum_{l=1}^{L} \|\mathbf{g}(l) - \sum_{k=1}^{K_{spc}} \gamma(l)_k \mathbf{a}(z_k)\|_2^2$

    $c \leftarrow c + 1$

  until $|\mu_c - \mu_{c-1}| \le \zeta$

Output: $\{\hat{z}_k\}$, $\hat{\mathbf{P}}$

IV. Numerical Experiments

Extensive numerical experiments have been conducted for evaluating the performance of the proposed DIARA. Simulated P-band SAR data are used to assess elevation estimation accuracy and resolution. In accordance with the Rayleigh resolution requirements and aircraft fuel constraints, eight acquisitions represent the upper limit of a single flight, while four acquisitions define the minimum for separating ground and canopy scatterers [12]. Experiments are performed using both P-band and L-band microstacks, where tomographic profiles reconstructed by DIARA are compared against reference LiDAR measurements. Four representative methods, i.e., Capon [28], MUSIC [35], RIAA [31], and WCS [40], are evaluated and compared in terms of reconstruction accuracy, resolution, and scatterer separability across different acquisition scenarios.

A. Performance Assessment Using Simulated Data

The parameters for the radar system applied in the experiments are summarized in Table I. To simulate varying elevation

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TABLE I
PARAMETERS OF THE SIMULATED DATA

Parameters	Value	Units
Carrier frequency	450	MHz
Reference distance	6500	m
SNR	5	dB
Baseline interval	30	m
A number of acquisitions (MBC1)	8	-
Maximum baseline (MBC1)	210	m
A number of acquisitions (MBC2)	4	-
Maximum baseline (MBC2)	90	m

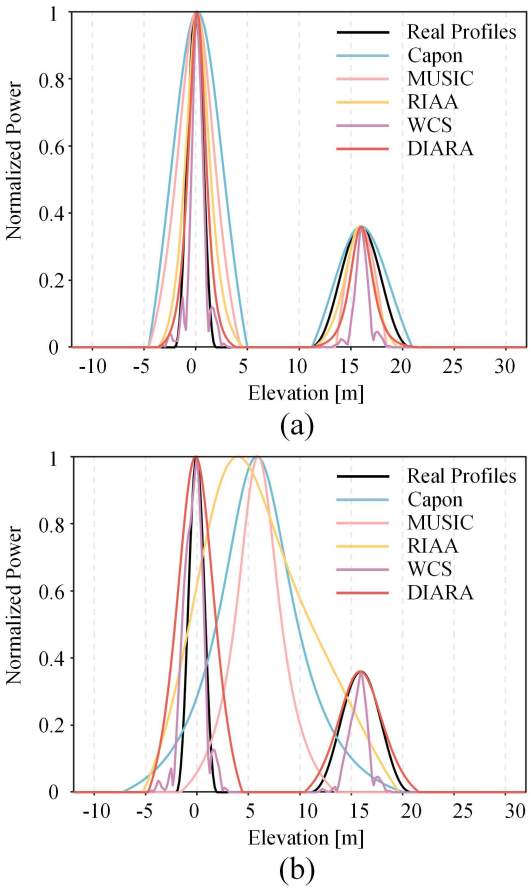


Fig. 4. Reconstructed tomographic reflectivity profiles along the elevation direction under different multibaseline configurations. (a) Results obtained using MBC1 (eight acquisitions). (b) Results obtained using MBC2 (four acquisitions). Five methods are compared: Capon, MUSIC, RIAA, WCS, and the proposed DIARA.

aperture conditions, the number of acquisitions is set to eight for the full-stack case and four for the microstack configuration. Two multibaseline configurations, referred to as MBC1 and MBC2, with baseline extents of 210 and 90 m, respectively, are considered. The signal-to-noise ratio (SNR) is fixed at 5 dB for all experiments. The reconstruction performance of the proposed DIARA framework is quantitatively assessed in terms of estimation accuracy, robustness to noise, and super-resolution capability.

1) Accuracy and Robustness: The simulated signal contains ground and canopy scatterers with distinct phase centers, located at different elevations and exhibiting varying angular

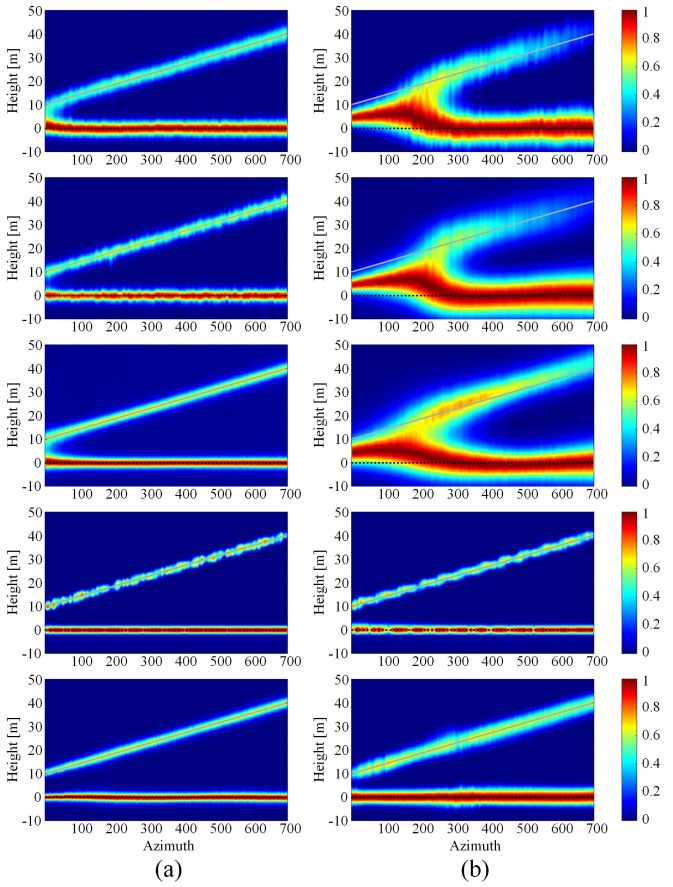


Fig. 5. Tomographic profiles with (a) MBC1 and (b) MBC2 by the five methods. (Top to bottom) Capon, MUSIC, RIAA, WCS, and DIARA, respectively. The upper white solid line indicates the canopy position, while the lower black dotted line shows the ground position.

spreads. As shown in Fig. 4, the canopy scatterer exhibits broader angular dispersion compared to the ground scatterer, consistent, with a distance of 16 m between the two SPCs. The reconstructed tomographic profiles of the Capon, MUSIC, RIAA, WCS, and DIARA method are presented in Fig. 4, respectively. The results are derived using an 11×11 pixel sliding window for covariance matrix estimation. Under the MBC1 configuration, all methods possess the capability to resolve the ground and canopy scatterers accurately. However, when using the reduced-aperture MBC2 configuration, Capon, MUSIC, and RIAA fail to separate the two scatterers, whereas WCS and DIARA still achieve reliable reconstruction, highlighting their superior performance under microstack conditions.

To provide a more comprehensive validation, a 2-D tomogram simulation was performed across a range of canopy heights. The scene consists of a ground plane at 0 m and a canopy layer whose height varies from 10 to 40 m along the azimuth direction. The distance between the two SPCs is varied correspondingly, which simulates forest scenarios of different canopy heights.

The 2-D tomograms reconstructed by the five methods are presented in Fig. 5 for both MBC1 and MBC2 configurations. Under MBC1 configuration, all methods successfully

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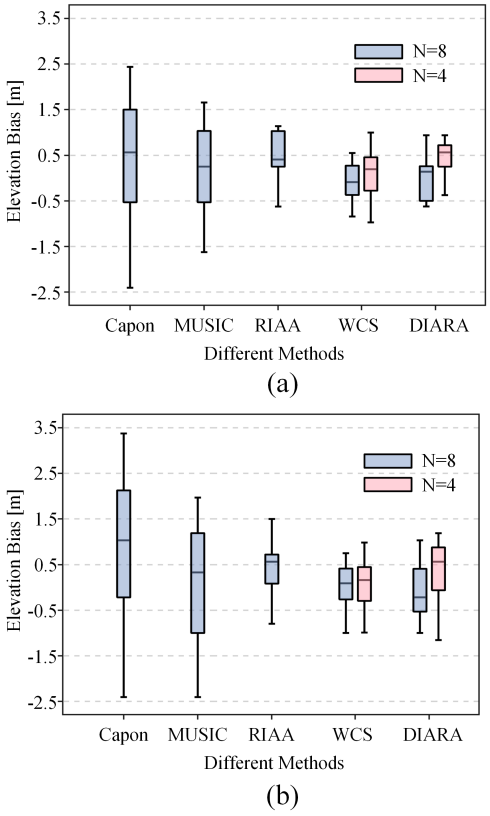


Fig. 6. Boxplot comparison of elevation estimation bias for five methods under different SNR conditions. (a) SNR = 10 dB. (b) SNR = 5 dB. The box boundaries indicate the first and third quartiles, the central bar denotes the median, and the whiskers represent the minimum and maximum values.

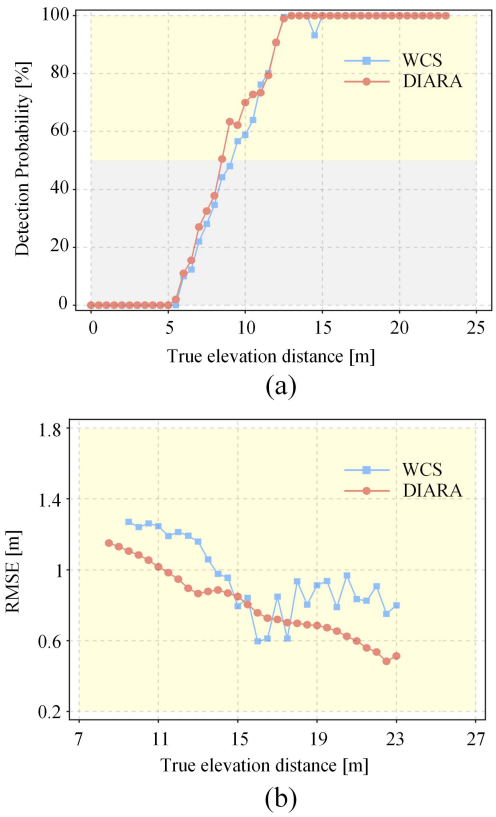


Fig. 7. P_D and RMSE versus vertical separation distance for the MBC2. (a) P_D values, with the yellow region showing P_D above 50%. (b) RMSE of elevation estimates.

reconstruct the ground and canopy layers, with the white solid line indicating the canopy height and the black dotted line showing the ground position. For the more challenging microstack MBC2 configuration, a clear degradation in performance is observed for all methods. However, the proposed DIARA and WCS maintain a clear separation between the ground and canopy layers, confirming their robustness to limited data. Specifically, the Capon and MUSIC tomograms show substantial blurring, with the ground and canopy layers merging, especially in areas with smaller canopy heights. RIAA shows slightly better performance but still fails to resolve the two layers consistently. In particular, the tomogram using DIARA exhibits high fidelity and stability across the entire range of canopy heights, providing a more reliable representation of FVS.

To further assess robustness, the elevation estimation accuracy is analyzed under SNR values of 10 and 5 dB, with the positional bias of the retrieved scatterers serving as the assessment criterion. For each SNR condition, 2000 Monte Carlo simulations are conducted to ensure the statistical reliability. As shown in Fig. 6(a), under 10-dB SNR, WCS and DIARA exhibit more consistent estimation with smaller bias fluctuations compared with Capon, MUSIC, and RIAA, even with only four acquisitions. Notably, DIARA achieves better error stability than WCS under the same acquisition conditions. Under 5-dB SNR [see Fig. 6(b)], the bias variance for Capon, MUSIC, and RIAA increases significantly, indicating high noise sensitivity, while WCS and DIARA maintain stable estimation accuracy. With four acquisitions, DIARA achieves comparable or better performance than WCS, confirming its robustness in low-SNR environments. These results demonstrate that DIARA effectively mitigates elevation estimation errors and provides reliable reconstruction performance under noisy and data-limited conditions.

2) Super-Resolution Capability: To assess the super-resolution capability of the DIARA, its performance is compared with that of the WCS method under a microstack configuration comprising four SAR acquisitions. The Rayleigh resolution in this setting is given by $\rho_h = \lambda R/(2B) = 23.8$ m. The simulations assess the detection probability (P_D) and RMSE of elevation estimates for varying distances between two SPCs, ranging from 0 to 23 m. The results, obtained from 2000 Monte Carlo simulations, are illustrated in Fig. 7. Fig. 7(a) shows the detection probability, where the yellow region indicates $P_D > 50\%$, and the gray region represents $P_D \leq 50\%$. Following the criterion defined in [50] and [52], the elevation resolution is determined as the minimum vertical separation corresponding to $P_D = 50\%$. Under this criterion, WCS achieves a resolution of 9.5 m, whereas DIARA resolves separations as small as 8.5 m, indicating that both methods surpass the Rayleigh limit. Fig. 7(b) further presents the RMSE curves for $P_D = 50\%$, showing that the elevation accuracy of DIARA improves as the phase center distance increases. In summary, the results demonstrate that DIARA achieves superior estimation accuracy compared to Capon, MUSIC, and

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TABLE II

TIME COSTS OF FIVE METHODS FOR TWO MULTIBASELINE CONFIGURATIONS (INTEL CORE I5-12500H CPU)

Configurations \ Methods	Capon	MUSIC	RIAA	WCS	DIARA
MBC1, $N = 8$	3.72 s	3.55 s	6.66 s	412.16 s	**26.01 s**
MBC2, $N = 4$	3.23 s	2.82 s	4.51 s	356.21 s	**35.79 s**

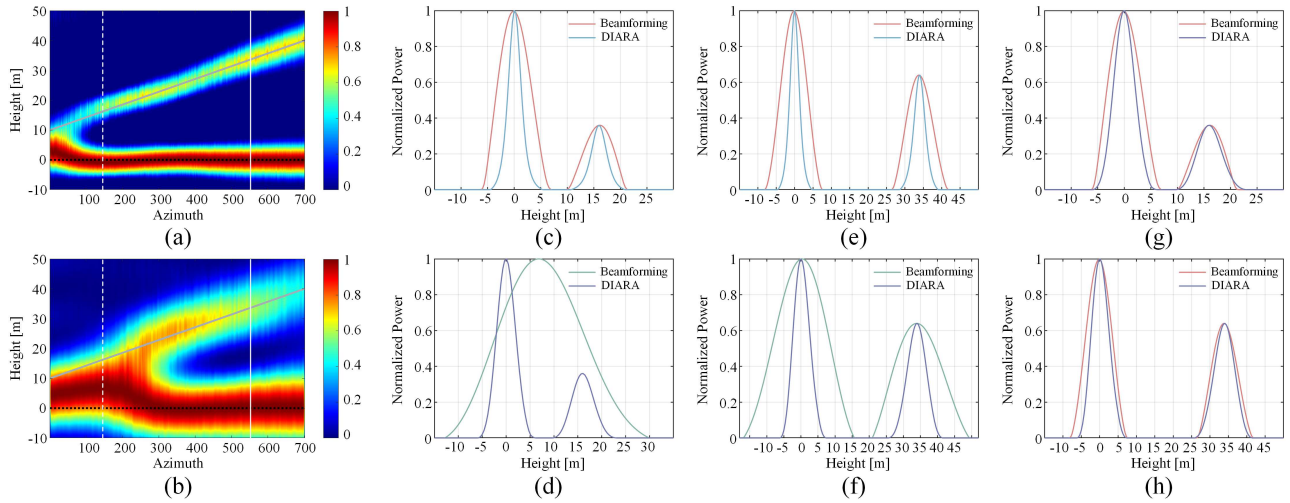


Fig. 8. Comparison of reflectivity profiles using beamforming and DIARA for two different baseline configurations. (a) and (b) Beamforming tomograms reconstructed from MBC1 and MBC2, respectively. (c)–(f) Reflectivity profiles along the white dashed and solid lines indicated in (a) and (b), comparing the results of beamforming and DIARA for the same acquisitions. (g) and (h) Cross comparison of DIARA result from MBC2 against the beamforming result from MBC1 for the profiles along the dashed and solid lines, respectively.

RIAA, while maintaining performance comparable to WCS. Notably, DIARA retains its super-resolution and robustness capabilities with only four acquisitions.

3) Computational Complexity: The computational cost of each method was evaluated by examining its principal operations. For the Capon estimator, the dominant workload arises from inverting the data covariance matrix. The runtime of MUSIC is determined by traversing the eigenvalue decomposition of that same covariance matrix to separate signal and noise subspaces. In RIAA, repeated matrix inversions within its iterative loop constitute the primary computational effort, and the total runtime increases with the number of iterations required for convergence. For WCS, the reconstruction is framed as a regularization-based optimization problem that is solved through numerous iterations, resulting in substantial computational demands. The processing burden of the DIARA is attributed to a two-tiered iterative scheme that includes an inner adaptation and an outer residual optimization, executing matrix inversions until the convergence criteria are satisfied.

Table II presents the average execution times measured on an Intel Core i5-12500H CPU (2.5 GHz, 16-GB RAM). As anticipated, the Capon, MUSIC, and RIAA execute more rapidly than the WCS and DIARA, confirming the greater efficiency of conventional spectral estimators. However, DIARA requires significantly less time than WCS, which highlights a favorable balance between imaging performance and processing efficiency.

Notably, reducing the number of acquisitions from 8 to 4 decreases the computation time of the Capon, MUSIC, RIAA, and WCS, due to smaller matrix dimensions. In contrast, the DIARA's runtime increases under the same conditions. The result stems from its iterative framework, where fewer baselines yield a less constrained optimization problem, which in turn extends the overall computation time required for convergence.

4) Radiometric Accuracy: The radiometric accuracy of TomoSAR is a crucial metric, particularly when dealing with distributed scattering scenarios [8]. While conventional beamforming effectively preserves radiometric accuracy for retrieving distributed scatterer power distributions, it is inherently limited in its vertical resolution [25]. In contrast, high-resolution methods such as Capon, MUSIC, RIAA, and WCS typically sacrifice the radiometric accuracy to achieve superior vertical resolution [33], [53].

To evaluate the radiometric performance of DIARA, we use the reflectivity profiles generated by beamforming with the MBC1 configuration as a reference. A comprehensive comparison of the reflectivity profiles reconstructed by both beamforming and DIARA under two multibaseline configurations (MBC1 and MBC2) is presented in Fig. 8. As shown in Fig. 8(a) and (b), beamforming fails to separate the two scatterers under microstack conditions. Fig. 8(c)–(f) demonstrates that when comparing results from the same baseline configuration, DIARA consistently produces a sharper main lobe, indicating a higher vertical resolution compared with beamforming. This gain in resolution, however, comes with a slight decrease in radiometric accuracy, which is consistent with the general compromise observed in other high-resolution methods.

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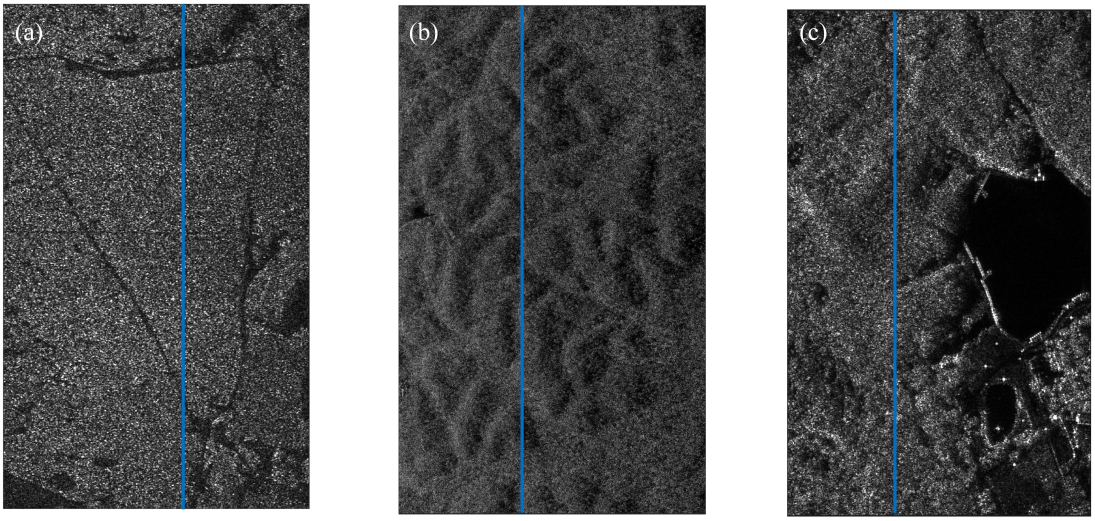


Fig. 9. SAR images of the study area. (a) Saihanba site. (b) Paracou site. (c) Diaoluoshan site. The blue line denotes the tomographic profile position.

TABLE III
RADIOMETRIC ACCURACY OF THE DIARA

Configurations	RMSE (dB)	MAE (dB)
MBC1, $N = 8$	8.53	6.37
MBC2, $N = 4$	5.89	4.64

An unexpected finding is highlighted in Fig. 8(g) and (h), which compare the DIARA result obtained with the sparser MBC2 configuration with the beamforming result obtained with MBC1. Interestingly, the reflectivity profiles of DIARA with only four images align well with those of beamforming using all eight images. The results suggest that the double iterative mechanism of DIARA not only significantly enhances resolution but also effectively preserves radiometric fidelity, even with a limited number of baselines. This capability represents a critical advantage, demonstrating DIARA's potential to achieve high-quality tomographic products with microstack configurations.

To quantitatively substantiate this observation, we have computed the RMSE and MAE of the DIARA profiles relative to the beamforming profiles from MBC1, as summarized in Table III. The results for the MBC2 configuration show that DIARA achieves an RMSE of 5.89 dB and an MAE of 4.64 dB, values that are notably lower than those obtained for the MBC1 configuration. The quantitative results corroborate that DIARA, even with fewer baselines, can produce results with a high degree of radiometric consistency with the baseline beamforming method. This unique capability positions DIARA as a highly efficient and robust method for TomoSAR applications where the availability of a large number of acquisitions is limited.

B. Performance Analysis Using BorTomoSAR P-Band TIS

1) Data Description: The TIS of the BorTomoSAR were acquired using a full-polarization P-band SAR system during an airborne field campaign conducted in Hebei, China, in October 2019. The SAR data were acquired using a Cessna aircraft as the platform over the Saihanba forest site. Due to constraints related to Rayleigh resolution, fuel capacity, and flight duration, a maximum of eight SAR acquisitions was obtained in a single flight, which we refer to as the multistack experiment (MuSE) [9]. This configuration provides a comprehensive evaluation of the airborne P-band TomoSAR system. To assess the methods under more constrained acquisition conditions, a reduced set of four SAR images was also used. This configuration represents the minimum number of acquisitions required for TomoSAR imaging to distinguish ground and canopy scatterers [12] and is referred to as the microstack experiment (MSE). For performance validation, both configurations are compared with LiDAR-derived canopy height model (CHM). Due to limitations in LiDAR coverage, only a portion of the TIS was selected as the experimental area. The system parameters used in both configurations are summarized in Table IV.

2) Multistack Tomographic Reflectivity Profiles: In the MuSE, eight SAR acquisitions formed a vertical baseline span of 210 m, resulting in a Rayleigh resolution of 10.3 m. A representative tomographic profile was extracted from the 200th range cell, as indicated by the blue line in Fig. 9(a). In forested scenarios, it is typically sufficient to assume $K_{\text{spc}} = 2$, corresponding to dominant the ground and canopy scatterers. Tomograms of HH and HV polarizations are reconstructed using Capon, MUSIC, RIAA, WCS, and the proposed DIARA methods, which are presented in Fig. 10. The LiDAR-derived CHM is superimposed as a white solid line for visual comparison. All five methods reveal a distinguishable FVS, as the tree height exceeds the system's vertical resolution. The HH polarization primarily captures strong ground scattering, while HV is more sensitive to canopy volume scatterers. Among the five methods, Capon and MUSIC produce broader and noisier responses. The RIAA performs moderately well, though some structural discontinuities remain visible. The WCS shows improved performance compared with classical methods. In contrast, the DIARA consistently produces tomograms with higher focusing quality, suppressed sidelobes, and continuous elevation structures across all three polarizations. Notably, the

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TABLE IV
DATA INFORMATION

Parameter name	BorTomoSAR	TropiSAR	HaiTomoSAR
Band	P-band	P-band	L-band
Carrier Frequency	450 MHz	397.5 MHz	1.25 GHz
Center Incidence Angle	60°	45°	50°
Flight Heading	North-South	South-North	East-West
Polarization	HH and HV	HH and HV	HH and HV
SLC Resolution (a×r)	1.45 m × 1.50 m	1.5 m × 1.2 m	0.66 m × 0.68 m
Vertical baseline length	90 m	45 m	30 m
Baseline Interval	30 m	15 m	10 m
Multi-stack	8	-	-
Micro-stack	4	4	4
Forest types	Temperate forest	Tropical forest	Tropical forest

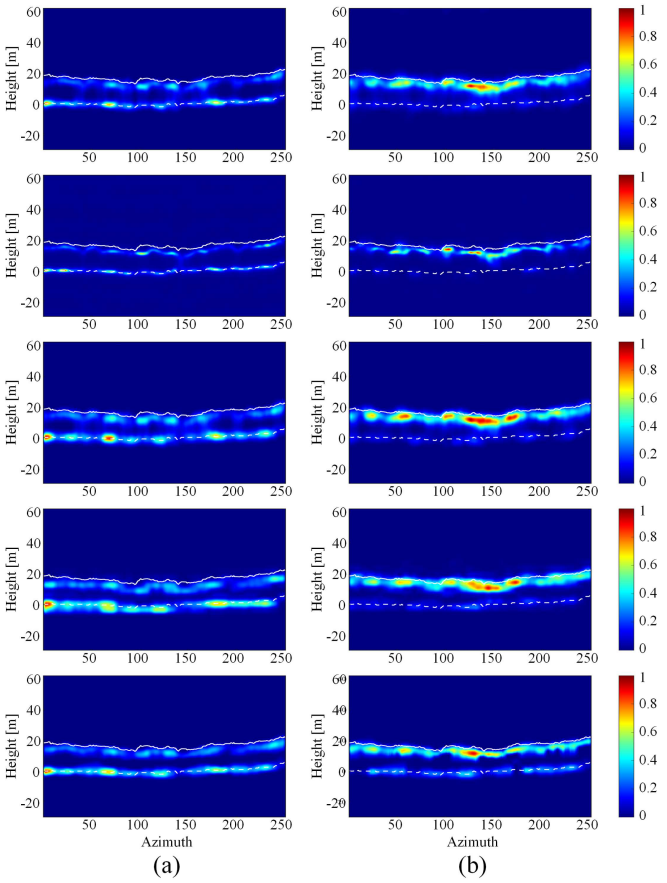


Fig. 10. Tomographic profiles of the Saihanba site by five methods using eight images. (a) HH polarization. (b) HV polarization. (Top to bottom) Profiles correspond to Capon, MUSIC, RIAA, WCS, and DIARA, respectively. The upper white solid line indicates the tree top position from LiDAR data, while the lower white dotted line shows the ground position from the DIARA.

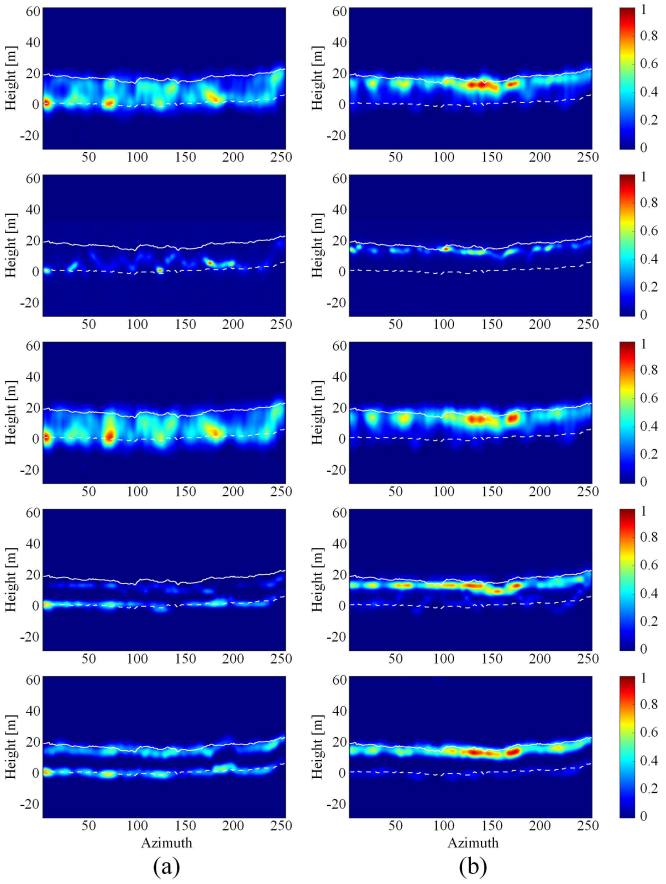


Fig. 11. Tomographic profiles of the Saihanba site by five methods using four images. (a) HH polarization. (b) HV polarization. (Top to bottom) Profiles correspond to Capon, MUSIC, RIAA, WCS, and DIARA, respectively. The upper white solid line indicates the tree top position from LiDAR data, while the lower white dotted line shows the ground position from the DIARA.

DIARA tomograms closely match the LiDAR CHM, which is superimposed as a white solid line for direct comparison. The results confirm the DIARA's superior reflectivity reconstruction compared with the WCS method.

3) Microstack Performance: To further evaluate the robustness of DIARA under constrained acquisition conditions, MSE was conducted using only four SAR images. This configuration provides a vertical baseline of 90 m and a Rayleigh resolution of 24.1 m, which is approximately twice that of MuSE. Tomograms are performed at the same spatial positions as in the MuSE. The reflectivity profiles obtained by the five methods under HH and HV polarizations are shown in Fig. 11. Due to the reduced number of acquisitions, Capon, MUSIC, and RIAA exhibit clear limitations, as seen in the first-to-third rows of Fig. 11. Their tomograms show broadened mainlobes and significant mixing between ground and canopy scatterers,

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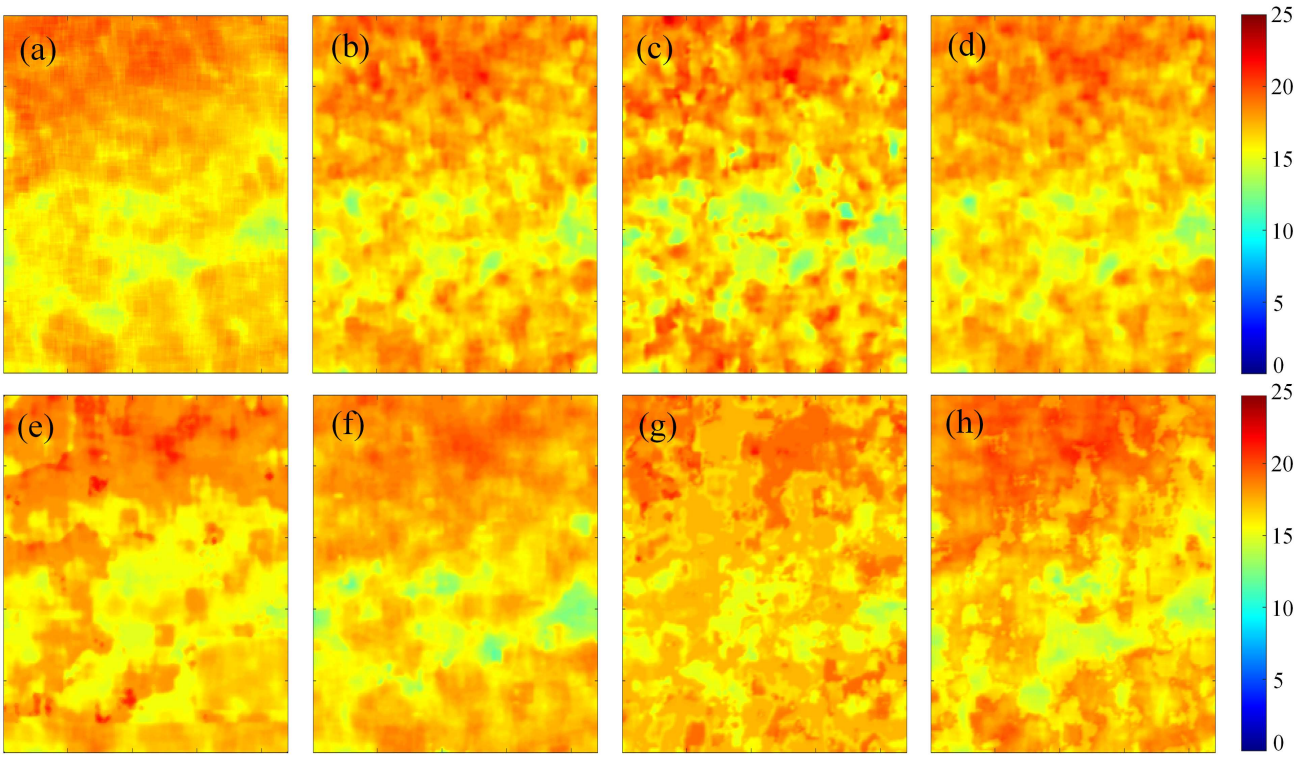


Fig. 12. LiDAR CHM and the forest height estimated by five methods. (a) LiDAR CHM, (b) Capon, (c) MUSIC, (d) RIAA, (e) WCS, and (f) DIARA with $N = 8$. (g) WCS, and (h) DIARA with $N = 4$, respectively.

resulting in poor vertical separability. In contrast, both WCS and DIARA are capable of distinguishing the ground and canopy scatterers with only four images. However, the WCS method relies on tuned hyperparameters, which may need to be adapted for different forest types or canopy densities (fourth row of Fig. 11). The DIARA, by contrast, produces well-focused and vertically continuous tomograms without the need for parameter adjustment. Its profiles remain smooth, coherent, and structurally consistent across all polarizations.

Notably, the tomograms reconstructed by DIARA under the MSE configuration remain qualitatively consistent with those obtained in the MuSE. This qualitative similarity highlights both the strong penetration capability of P-band SAR and the robustness of DIARA in handling reduced-baseline TIS. The preservation of ground–canopy separability and structural detail, even under acquisition-limited conditions, confirms the adaptability and reliability of DIARA for airborne TomoSAR applications where the number of passes may be constrained.

4) Comparison of Retrieved Forest Heights: Forest height estimation is performed by analyzing the vertical reflectivity profiles and their corresponding power distribution [7], [13]. In the MuSE, all five methods demonstrate reasonable capability in forest structure reconstruction. However, under MSE with only four acquisitions, Capon, MUSIC, and RIAA fail to generate sufficiently accurate tomograms. Consequently, forest height estimation in the MSE is limited to the WCS and DIARA methods. As shown in Figs. 10 and 11, HH polarization predominantly captures ground backscatter and HV captures canopy volume scattering. Based on this, terrain height is estimated from HH polarization data, and canopy top height is derived from the HV polarization data. The absolute canopy height is then computed as the vertical distance between the peak backscatter responses of the canopy and ground scatterers. By applying power loss thresholds ranging from 0 to −5 dB, the extracted canopy top height is compared with the LiDAR-derived CHM to assess accuracy. In the MuSE, a power loss threshold of −3 dB is identified as the optimal value for all methods, consistently yielding the lowest bias and RMSE for forest height estimation. Under the more constrained MSE, DIARA achieves optimal performance at a threshold of −3.5 dB, while WCS is optimal at −4 dB. The minor discrepancy in optimal thresholds is attributed to the inherent differences in their estimation of the ground and canopy SPCs. The sensitivity of height retrieval accuracy to this parameter highlights the importance of selecting an appropriate power loss threshold for different methods and acquisition scenarios. A visual comparison of forest height maps generated by the five methods, along with the LiDAR CHM, is provided in Fig. 12, while Fig. 13 presents the joint distribution between estimated heights and LiDAR CHM, indicating height ranges between 12 and 22 m.

To further evaluate accuracy and quantify performance, we compute the R^2, MAE, MAPE, and RMSE metrics relative to the LiDAR CHM (see Fig. 11). A rose diagram [see Fig. 14(a)] and normalized Taylor diagrams [see Fig. 14(b)] offer a comprehensive visual comparison. In addition, the DIARA method achieves high accuracy in the MuSE with R^2 values exceeding 0.7 and lower RMSE than the other methods,

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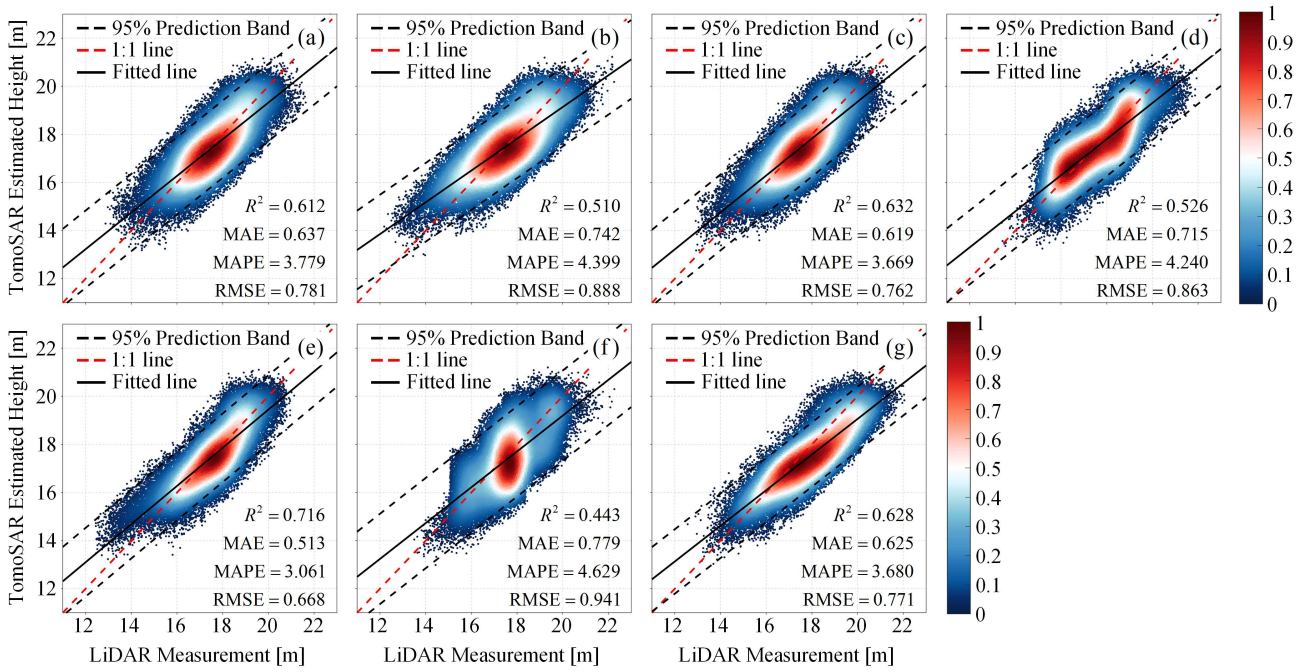


Fig. 13. Joint distribution between LiDAR CHM and forest height estimates produced by five methods. (a) Capon, (b) MUSIC, (c) RIAA, (d) WCS, and (e) DIARA with $N = 8$. (f) WCS and (g) DIARA with $N = 4$.

demonstrating superior consistency with LiDAR CHM. In the Taylor diagram, the DIARA (represented as green inverted triangles) is positioned closest to the reference point (blue circle), reflecting its superior balance of correlation, standard deviation, and error. Interestingly, the DIARA also achieves higher correlation in the MSE than in MuSE, highlighting its robustness under limited-TIS conditions. Overall, the results confirm that the DIARA outperforms classical nonparametric, parametric, and sparse estimations in forest height retrieval, particularly when limited SAR acquisitions are available. Its performance remains stable across different acquisition scenarios, validating its effectiveness for airborne TomoSAR applications. These results demonstrate the superiority of DIARA in high-resolution TomoSAR imaging, not only in terms of visual interpretability but also in quantitative accuracy and structural stability.

C. Performance Analysis Using TropiSAR P-Band TIS

1) Data Description: To demonstrate the robustness of the proposed DIARA method, we utilized four P-band SAR images from the TropiSAR 2009 campaign. The data were acquired over the forest site of Paracou, French Guiana [54]. The forest height in this area ranges from 20 to 40 m, which is both higher and more structurally complex than the previous study area. This configuration, with its high vegetation density and complex structure, provides a more rigorous test of the DIARA's capability. The system parameters for this experiment are presented in Table IV, and the SAR amplitude image of the study region is shown in Fig. 9(b).

2) Reflectivity Profiles From Four Images: A representative tomographic profile was extracted from a specific range cell to evaluate the performance of all five methods under this challenging condition. The tomograms incorporate LiDAR-derived profiles for spatial reference, depicting the digital terrain model (DTM) and digital surface model (DSM) with solid black and solid white lines, respectively. As shown in Fig. 15, the reconstructed tomograms for Capon, MUSIC, and RIAA struggle to provide a clear and well-focused vertical profile. Due to the limited number of baselines and the high density of the canopy, their mainlobes are broadened, and the profiles are noisy. While some degree of separation between the ground and canopy scatterers can be observed in certain areas, the overall performance is compromised, resulting in a significantly lower vertical separability compared with WCS and DIARA. The regularization-based method of WCS is capable of achieving super-resolution, its performance is highly sensitive to the choice of predefined parameters. An improper selection of the parameters can lead to undesirable signal artifacts and leakage, as evidenced in the fourth row of Fig. 15. In contrast, DIARA consistently yields tomograms with superior focusing and sidelobe suppression, thereby maintaining the structural continuity of the forest profile.

Notably, the tomograms of the proposed DIARA clearly resolve the distinct returns from both the dense canopy and the underlying ground. This ability to accurately penetrate and reconstruct the vertical structure of a complex and dense tropical forest further validates the robustness and effectiveness of our residual-based optimization strategy.

3) Comparison of Retrieved Forest Heights: The superior structural clarity of the DIARA tomograms translates into more accurate forest height estimates. The minimum bias and RMSE are achieved at a power loss of approximately −3 dB. As shown in the forest height map in Fig. 16, both the DIARA method and the reference LiDAR CHM are presented. We have chosen to only display the DIARA results, as the Capon, MUSIC, and RIAA fail to generate sufficiently complete

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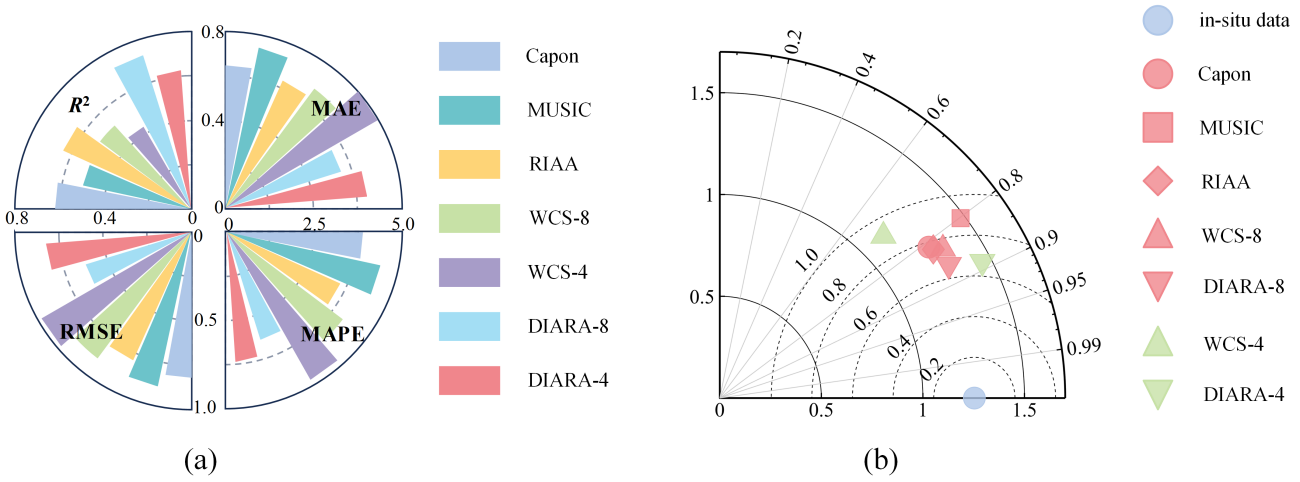


Fig. 14. Performance of five TomoSAR methods. (a) Performance indicators (R^2, MAE, MAPE, and RMSE). (b) Taylor diagram, with the radial dimension indicating the standard deviation normalized by the TomoSAR methods. The correlation coefficient is depicted in the angular coordinates, and RMSE is shown by the arcs. The blue circle marks CHM reference points. The WCS-8, DIARA-8, WCS-4, and DIARA-4 stand for data stacks of 8, 8, 4, and 4, respectively.

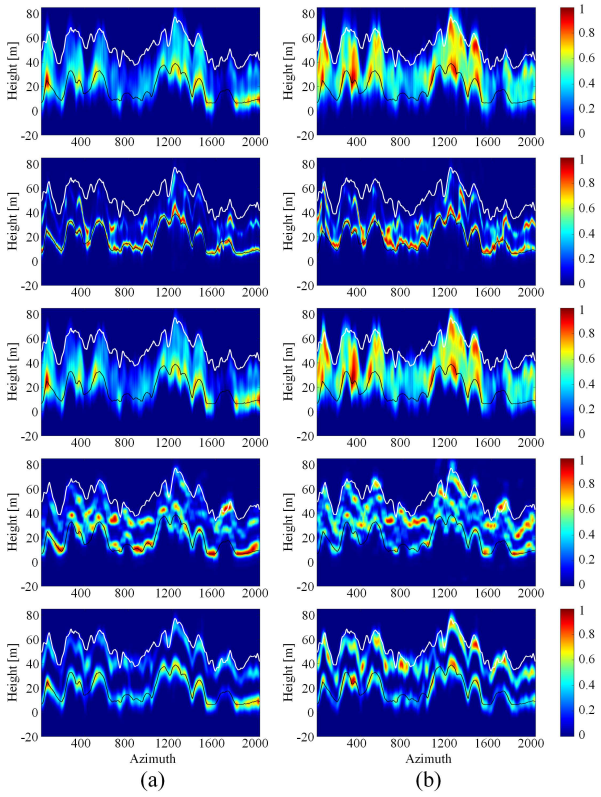


Fig. 15. Tomographic Profiles of the Paracou site by five methods using four images. (a) HH polarization. (b) HV polarization. (Top to bottom) Profiles correspond to Capon, MUSIC, RIAA, WCS, and DIARA. The upper white solid line and the lower black solid line indicate the LiDAR-derived DSM and DTM, respectively.

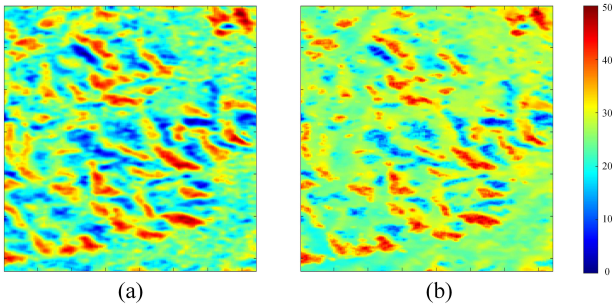


Fig. 16. Comparison of forest height extraction at the Paracou site. (a) LiDAR CHM and (b) DIARA TomoSAR method.

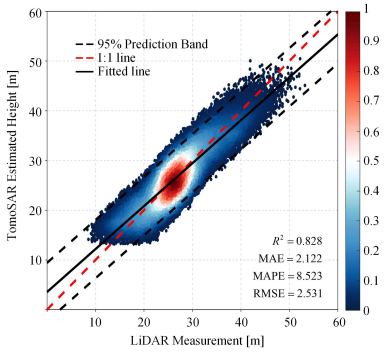


Fig. 17. Joint distribution of LiDAR CHM and the forest height estimated by the DIARA TomoSAR method at the Paracou site.

height maps due to their inability to consistently separate the ground and canopy signals in this challenging environment. Furthermore, while the WCS method is able to produce a height map, it was characterized by numerous artifacts and lower accuracy. Consequently, the visual comparison in Fig. 16 focuses on the most robust result from DIARA, which demonstrates excellent structural consistency with LiDAR CHM [55].

For a quantitative assessment, the joint distribution between the estimated heights and the LiDAR CHM is presented in Fig. 17, along with key evaluation metrics. The results show that DIARA achieves a stronger correlation and lower RMSE compared with WCS. The superior performance in this benchmark dataset, which represents a more complex and high-canopy scenario, further confirms the robustness and generalizability of the proposed DIARA framework. Its ability to accurately retrieve forest height in such a demanding

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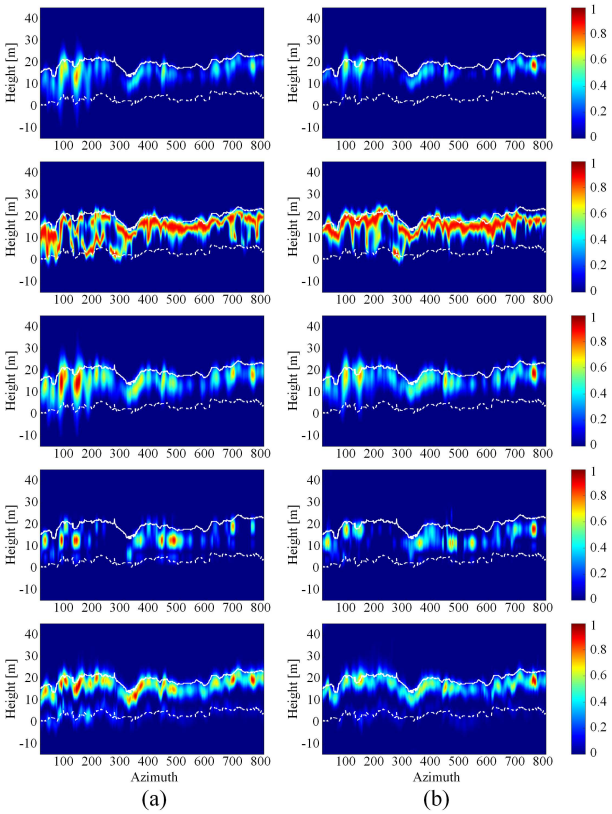


Fig. 18. Tomographic Profiles of the Diaoluoshan site by five methods using four images. (a) HH polarization. (b) HV polarization. (Top to bottom) Profiles correspond to Capon, MUSIC, RIAA, WCS, and DIARA. The upper white solid line indicates the tree top position from LiDAR CHM, while the lower white dotted line shows the ground position from the DIARA.

environment highlights its potential for large-scale forest mapping estimation.

D. Performance Analysis Using HaiTomoSAR L-Band TIS

1) Data Description: An airborne L-band tomographic SAR experiment, referred to as the HaiTomoSAR, was carried out on March 6, 2021, using a fully polarimetric system mounted on a Cessna aircraft. The data acquisition campaign was conducted over the Diaoluoshan National Nature Reserve in Hainan, China, which is known for its ecological diversity and serves as a representative tropical forest ecosystem. During the flight, the aircraft maintained an altitude between 4850 and 4880 m. The TIS consists of four fully polarimetric L-band SAR images acquired in repeat-pass mode over the same area. System parameters are summarized in Table IV, and the SAR amplitude image of the study region is shown in Fig. 9(c). The tropical forest environment provides a challenging testbed for tomographic reconstruction due to its dense vegetation, complex FVS, and strong volume scattering [56], [57]. These conditions make the MSE particularly well-suited for evaluating the robustness of the proposed DIARA method under limited baseline diversity.

2) Reflectivity Profiles From Four Images: The spatial baseline span of the L-band microstack is 30 m in the elevation direction, yielding a vertical Rayleigh resolution of 22.69 m.

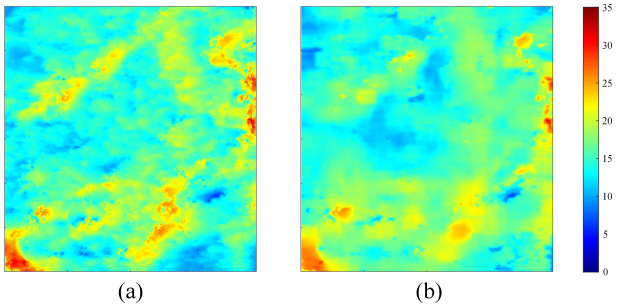


Fig. 19. Comparison of forest height extraction at the Diaoluoshan site. (a) LiDAR CHM and (b) DIARA TomoSAR method.

In this experiment, a specific range line, which is marked by the blue line in Fig. 9(b), is selected to evaluate and compare the performance of Capon, MUSIC, RIAA, WCS, and the proposed DIARA methods. The reconstructed tomograms are shown in Fig. 18, where the top white line represents LiDAR CHM. Notably, the dominant backscattering originates from the canopy scatterers, while ground backscatter is significantly attenuated due to limited penetration of the L-band SAR signal through the dense tropical canopy. In all polarizations, the tomographic profiles from Capon, MUSIC, RIAA, and WCS methods exhibit broadened mainlobes and high sidelobes (shown in the first to fourth rows of Fig. 18). However, the DIARA employs a residual-based optimization strategy that iteratively subtracts dominant scattering components. This mechanism effectively mitigates off-peak energy leakage from strong volume scatterers, enhancing the visibility of weak ground returns and yielding well-focused tomograms with low sidelobe levels. As a result, the canopy and ground scatterers are more distinctly separated, even under L-band conditions where dense vegetation heavily attenuates ground signals. Moreover, the DIARA shows well-focused profiles with significantly suppressed sidelobes (shown in the fifth row of Fig. 18), thereby ensuring good agreement with LiDAR CHM. The results confirm the robustness of DIARA in L-band TomoSAR forest scenarios, where conventional methods struggle with low penetration and complex scattering. The DIARA's ability to recover structurally coherent profiles across all polarizations highlights its potential for generalization to other tropical forest environments and L-band SAR missions.

3) Comparison of Retrieved Forest Heights: Among the five methods, only the DIARA achieves sufficient TomoSAR focusing quality for forest height retrieval in the L-band experiment.An optimal power loss threshold of approximately −3.5 dB is identified, which minimized both bias and RMSE for canopy top height estimation. Fig. 19 presents the LiDAR CHM and the forest height retrieval from TomoSAR for the study area. Fig. 20 shows the joint distribution between the estimated forest height and the LiDAR CHM. It also provides the evaluation metrics for forest height estimation, including R^2, MAE, MAPE, and RMSE. The retrieved forest heights exhibit high structural consistency with LiDAR CHM across the study area, confirming its capacity to produce structurally consistent and application, even when limited

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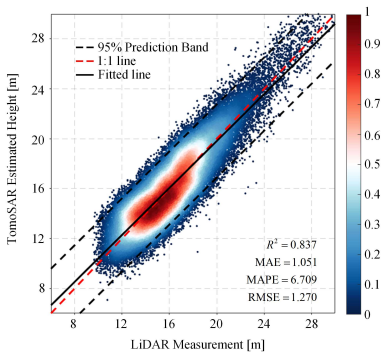


Fig. 20. Joint distribution of LiDAR CHM and the forest height estimated by the DIARA TomoSAR method at the Diaoluoshan site.

SAR acquisitions are available. Overall, the results confirm that DIARA provides accurate and structurally reliable forest height estimates in dense tropical environments, even under limited baseline conditions.

In conclusion, the DIARA method consistently achieves superior vertical resolution and structural clarity compared with traditional TomoSAR methods when applied to both P- and L-band SAR microstacks. The experimental results demonstrate its robustness across different frequency bands and vegetation types, highlighting its adaptability to constrained acquisition scenarios. By effectively suppressing sidelobes and enhancing weak ground returns through residual-based optimization, DIARA delivers accurate and structurally consistent forest height estimates even in dense tropical forests.

V. Conclusion

This article proposes a robust and efficient TomoSAR imaging method, termed the DIARA, tailored for ground–canopy scatterer separation and accurate forest height retrieval using a minimal number of SAR acquisitions. Unlike conventional methods, the DIARA combines inner–outer iterative adaptation with residual-based refinement, significantly improving accuracy and super-resolution capability. Tomographic focusing using inner IAA performs well with the limited elevation aperture size. Subsequently, the signal residual optimization approach addresses the coupling effect resulting from the ground and canopy signal interactions. The combined use of the double iterative thresholding and the residual refinement achieves the tomographic reflectivity profile with high resolution and low sidelobe.

Furthermore, the DIARA is validated in two distinct forest environments using only four SAR acquisitions to assess the quality of the reconstructed tomograms. In the P-band TomoSAR experiments over temperate forests, DIARA achieves better forest height estimation. It also shows sharper vertical profiles and more accurate ground-canopy separation than Capon, MUSIC, RIAA, and WCS. The TropiSAR P-band results further demonstrate the robustness of the proposed DIARA by successfully reconstructing vertical structures in complex forest environments. In L-band microstack TomoSAR experiment over dense tropical forests, it effectively detects weak ground signals and accurately recovers reflectivity profiles. The accuracy and reliability of the height retrievals are quantitatively confirmed via comparisons with LiDAR CHM. The results highlight that DIARA is not only adaptable across diverse forest types and bands but also has the potential to advance microstack and single-pass multibaseline SAR systems for efficient 3-D forest mapping with minimal data. In the future, the authors hope that the DIARA framework can be further validated on more spaceborne SAR platforms, such as the BIOMASS mission, and serve as a feasible solution for large-scale 3-D forest mapping with mirocstack TomoSAR.

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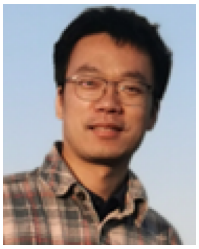
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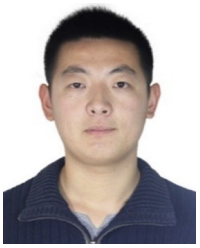
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